PeriGen, Inc.

PeriWatch Vigilance™ System Requirements

PeriGen
Advanced
Perinatal Systems

Table of Contents

Enterprise Model (Multi-Facilities) Server Requirements 3
PeriWatch Vigilance – Enterprise Application Server (VM) 3
*PeriWatch Vigilance – Mixed-Mode Data Acquisition VM (if required) 4
PeriWatch Vigilance - Test/Training server if required 4
Client / Workstation Requirements 4
Thin-Client Technologies 5
Network (For Workstations and Data Acquisition) 5
Tracing Data Acquisition 5
Fetal Monitor Compatibility 6

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

Enterprise Model (Multi-Facilities) Server Requirements

- Single server VMs providing the following main functions:
  - Application Services & Decision Support
  - SQL Server DB
  - HL7 ADT and results Interface
  - Report services

- System sizing is based on peak utilization and as a function of total monitored areas. Sizing below is based on projected requirements for approximately 5 years.
- Final configuration and optimization based on actual enterprise layout will be determined at system implementation planning stage
- VMware is currently supported on version 6.x. and above. Other platforms may require extra validation services
- NOTE: Mixed-mode enterprise data acquisition configurations across (example: One site using Network Redirectors while the other is using DIGI terminal servers), will require an additional small VM* per acquisition mode. The main enterprise server can accommodate one mode only.

PeriWatch Vigilance – Enterprise Application Server (VM)

Up to # of Monitored locations	50	100	200
# of vCPUs	4	6	8
GB RAM	12	16	20
Volume C: OS	100 GB	100 GB	100 GB
Volume D: Applications	40 GB	40 GB	40 GB
Volume E: SQLData	260 GB	330 GB	400 GB
Volume F: SQLTransactionLog	50 GB	50 GB	50 GB
Volume G : SQLBackup	260 GB	330 GB	400 GB

- Gigabit Ethernet (Number of NICs may change depending on load and storage considerations)
- Microsoft Windows Server 2012 R2
  - Internet Information Services (IIS) 8.0
  - ASP.NET
  - Windows Authentication
- MS SQL 2014 SP2 Standard (or above) with Reporting Services
- TCP communication ports: 80, 443, 8000-8020, UDP: 11002
- Windows server required modules on this and each server mentioned below:
  - Microsoft Web Services Enhancements (WSE) 3.0
  - VC++ 2010 SP1 Redistributable, VC++ 2013 Redistributable
  - .Net 3.5 SP1, .Net 4.7.1
  - .Net Core Windows Hosting 2.0.5

NOTE: Final sizing to be determined during pre-install implementation planning.

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

*PeriWatch Vigilance – Mixed-Mode Data Acquisition VM (if required)

- 2 vCPUs
- 6GB RAM
- Storage requirements per volume:
    - C: Operating system - 100 GB
- Gigabit Ethernet
- Microsoft Windows Server 2012 R2
- TCP: 80, 443, 8000-8020, UDP: 11002
- Windows server required modules as mentioned on the primary server

PeriWatch Vigilance - Test/Training server if required

- 4 vCPUs
- 8GB RAM
- Storage requirements per volume:
    - C: Operating system - 100 GB
    - D: Applications/SQLData/Logs - 200 GB
- Gigabit Ethernet
- Microsoft Windows Server 2012 R2
    - Internet Information Services (IIS) 8.0
    - ASP.NET
    - Windows Authentication
- MS SQL 2014 SP2 Standard (or above) with Reporting Services
- TCP communication ports: 80, 443, 8000-8020, UDP: 11002
  Windows server required modules as mentioned on the primary server

Client / Workstation Requirements

- Microsoft Windows 7/10 Professional 32/64-bit. 64 bit recommended.
- 4GB RAM. 500 MB of available memory should be reserved for PeriWatch Vigilance.
- 1024x768 (4:3) is the minimum supported resolution but it is not recommended
- 1280x1024 or higher wide screens resolutions are recommended
- Native resolutions only. non-native resolutions or aspect ratios will result in a stretched representation of fetal tracings and are not supported
- .NET Framework 4.7.1
- VC++ 2010 SP1 Redistributable, VC++ 2013 Redistributable
- Internet Explorer 11 web browser minimum
- MS Edge or Google Chrome recommended

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

Thin-Client Technologies

- PeriWatch Vigilance HUB is deployed as a .Net thick client supported on any thin-client technology (Citrix, VMware View).
- This version of the product was validated against Validation of this product of the specific thin-client infrastructure version used at the customer site will be required during system installation activities.
- Citrix server / VMware view server sizing should be aligned with PeriWatch Client requirements and number of concurrent session requirements (see previous section). Precise server and memory recommendations are a function of overall site load. Recommendations will be finalized during system implementation.

Network (For Workstations and Data Acquisition)

- Switched Ethernet network, 100 Mbit or better

Tracing Data Acquisition

PeriGen provide a proprietary, secure and resilient solution dedicated to collect data from existing terminal server hardware used by the existing surveillance system:

For sites using DIGI ConnectPort or PortServer hardware, PeriGen provide a Network Redirector (NR) hardware to mirror tracing network traffic. Network Redirector has the following requirements:

- POE capable switch. NR is a port powered device and need to be directly connected to a port where POE is enabled
- The NR must be connected to the same switch where the third-party DIGI device is connected
- NR must have two ports available on the said switch. One for receiving one for sending.
- Only the NR's receiving port requires POE. The sending port does not require POE and can be connected to another switch in case the primary switch is full.
- Customer needs to configure "port mirroring" from the port where the third-party system DIGI device is connected to the port where the NR is connected.

For sites using MOXA terminal server, com port mirroring must be enable on the MOXA device directly to allow a parallel data acquisition.

For other configurations, please contact PeriGen support at 1-855-236-4353, support@perigen.com

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.

Fetal Monitor Compatibility

Vendor	Protocol compatibility	Type	Interface card / Firmware	Description
Philips	1371	* 1350A and B * 50XM series	M1350-66532 or M1350-66536 * Firmware: **C01.03 and above**	DB9 pin serial connector for RS232 interface. Firmware software must support option C13.
Philips	1371	Avalon FM20 Avalon FM30 Avalon FM40 Avalon FM50	Firmware: **F.0X and above.**	DB9 pin serial connector for RS232 interface. Firmware software must support option C13.
GE/Corometrics	1371	118	J09-J10	DB9 RS232 serial interface
GE/Corometrics	1371	120/170/250 Series	J09-J10-J11	RJ11 RS232 serial interface

* Philips 1350 A/B and 50XM are presently out of support by Philips. Although the serial protocol is still supported, PeriGen cannot be held responsible for known issues with these fetal monitors with firmware versions lower than **C01.03.** Known issues include a very high frequency of SP02 measurements sent to the electronic tracing when the paper recorder door is open and no paper is in the-0 tray.

©PeriGen, Inc. – Confidential and Proprietary
This document may not be copied without express written permission.